

About

These pinouts show what pins are used as configured in the ArPiRobot OS images
These do not show all possible uses of each pin or all hw interfaces available on the board
See official pinouts to locate this (official documentation is linked for each board)

Raspberry Pi Zero W



	Physical	
3.3V	1 2	5V
I2C1 SDA	3 4	5V
I2C1 SCL	5 6	GND
GPIO 4	7 8	UART0 TX
GND	9 10	UART0 RX
GPIO 17	11 12	GPIO 18
GPIO 27	13 14	GND
GPIO 22	15 16	GPIO 23
3.3V	17 18	GPIO 24
SPI0 MOSI	19 20	GND
SPI0 MISO	21 22	GPIO 25
SPI0 SCLK	23 24	SPI0 CS0
GND	25 26	SPI0 CS1
I2C0 SDA	27 28	I2C0 SCL
GPIO 5	29 30	GND
GPIO 6	31 32	GPIO 12
GPIO 13	33 34	GND
GPIO 19	35 36	GPIO 16
GPIO 26	37 38	GPIO 20
GND	39 40	GPIO 21

RPi Hat Compatible	Yes – Full 40 pin
Rpi Camera Compatible	Yes – requires special cable
UART Console	UART0
UART Console Baud Rate	115200
Reserved Interfaces	UART0, I2C0
Number USB	1 x USB micro-B OTG
Powered By	USB micro-B
Boots From	micro SD Card
Built-in WiFi	Yes – 2.4GHz
Ethernet Port	No
CPU Cores	1
RAM	512MB
GPU Video Decoding	Yes

<https://www.raspberrypi.com/documentation/computers/ras>

Raspberry Pi Zero 2 W

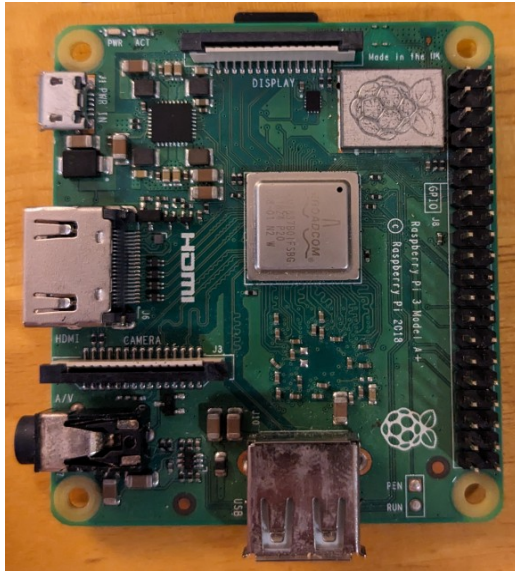


	Physical		
3.3V	1	2	5V
I2C1 SDA	3	4	5V
I2C1 SCL	5	6	GND
GPIO 4	7	8	UART0 TX
GND	9	10	UART0 RX
GPIO 17	11	12	GPIO 18
GPIO 27	13	14	GND
GPIO 22	15	16	GPIO 23
3.3V	17	18	GPIO 24
SPI0 MOSI	19	20	GND
SPI0 MISO	21	22	GPIO 25
SPI0 SCLK	23	24	SPI0 CS0
GND	25	26	SPI0 CS1
I2C0 SDA	27	28	I2C0 SCL
GPIO 5	29	30	GND
GPIO 6	31	32	GPIO 12
GPIO 13	33	34	GND
GPIO 19	35	36	GPIO 16
GPIO 26	37	38	GPIO 20
GND	39	40	GPIO 21

RPi Hat Compatible	Yes – Full 40 pin
Rpi Camera Compatible	Yes – requires special cable
UART Console	UART0
UART Console Baud Rate	115200
Reserved Interfaces	UART0, I2C0
Number USB	1 x USB micro-B OTG
Powered By	USB micro-B
Boots From	micro SD Card
Built-in WiFi	Yes – 2.4GHz
Ethernet Port	No
CPU Cores	4
RAM	512MB
GPU Video Decoding	Yes

<https://www.raspberrypi.com/documentation/computers/ras>

Raspberry Pi 3A+



	Physical	
3.3V	1 2	5V
I2C1 SDA	3 4	5V
I2C1 SCL	5 6	GND
GPIO 4	7 8	UART0 TX
GND	9 10	UART0 RX
GPIO 17	11 12	GPIO 18
GPIO 27	13 14	GND
GPIO 22	15 16	GPIO 23
3.3V	17 18	GPIO 24
SPI0 MOSI	19 20	GND
SPI0 MISO	21 22	GPIO 25
SPI0 SCLK	23 24	SPI0 CS0
GND	25 26	SPI0 CS1
I2C0 SDA	27 28	I2C0 SCL
GPIO 5	29 30	GND
GPIO 6	31 32	GPIO 12
GPIO 13	33 34	GND
GPIO 19	35 36	GPIO 16
GPIO 26	37 38	GPIO 20
GND	39 40	GPIO 21

RPi Hat Compatible Yes – Full 40 pin
Rpi Camera Compatible Yes

UART Console UART0
UART Console Baud Rate 115200
Reserved Interfaces UART0, I2C0

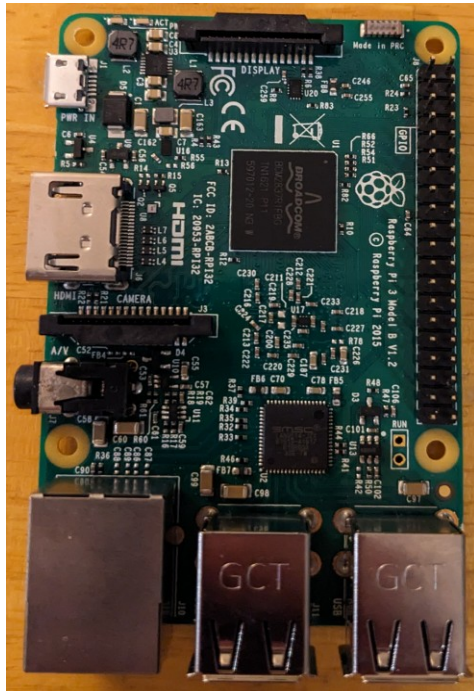
Number USB 1 x USB-A
Powered By USB micro-B
Boots From micro SD Card

Built-in WiFi Yes – 2.4GHz, 5.0GHz
Ethernet Port No

CPU Cores 4
RAM 512MB
GPU Video Decoding Yes

<https://www.raspberrypi.com/documentation/computers/ras>

Raspberry Pi 3B(+)



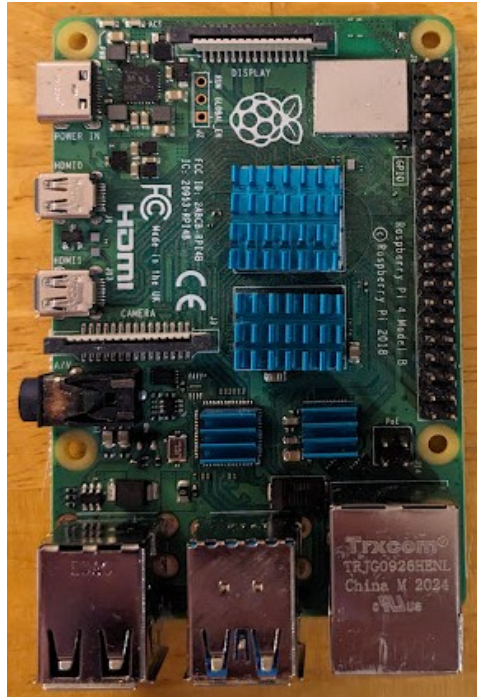
Physical

3.3V	1 2	5V
I2C1 SDA	3 4	5V
I2C1 SCL	5 6	GND
GPIO 4	7 8	UART0 TX
GND	9 10	UART0 RX
GPIO 17	11 12	GPIO 18
GPIO 27	13 14	GND
GPIO 22	15 16	GPIO 23
3.3V	17 18	GPIO 24
SPI0 MOSI	19 20	GND
SPI0 MISO	21 22	GPIO 25
SPI0 SCLK	23 24	SPI0 CS0
GND	25 26	SPI0 CS1
I2C0 SDA	27 28	I2C0 SCL
GPIO 5	29 30	GND
GPIO 6	31 32	GPIO 12
GPIO 13	33 34	GND
GPIO 19	35 36	GPIO 16
GPIO 26	37 38	GPIO 20
GND	39 40	GPIO 21

RPi Hat Compatible	Yes – Full 40 pin
Rpi Camera Compatible	Yes
UART Console	UART0
UART Console Baud Rate	115200
Reserved Interfaces	UART0, I2C0
Number USB	4 x USB-A
Powered By	USB micro-B
Boots From	micro SD Card
Built-in WiFi	Yes – 2.4GHz, 5.0GHz (5.0G
Ethernet Port	Yes
CPU Cores	4
RAM	1GB
GPU Video Decoding	Yes

<https://www.raspberrypi.com/documentation/computers/ras>

Raspberry Pi 4B



	Physical	
3.3V	1 2	5V
I2C1 SDA	3 4	5V
I2C1 SCL	5 6	GND
GPIO 4	7 8	UART0 TX
GND	9 10	UART0 RX
GPIO 17	11 12	GPIO 18
GPIO 27	13 14	GND
GPIO 22	15 16	GPIO 23
3.3V	17 18	GPIO 24
SPI0 MOSI	19 20	GND
SPI0 MISO	21 22	GPIO 25
SPI0 SCLK	23 24	SPI0 CS0
GND	25 26	SPI0 CS1
I2C0 SDA	27 28	I2C0 SCL
GPIO 5	29 30	GND
GPIO 6	31 32	GPIO 12
GPIO 13	33 34	GND
GPIO 19	35 36	GPIO 16
GPIO 26	37 38	GPIO 20
GND	39 40	GPIO 21

RPi Hat Compatible Yes – Full 40 pin
Rpi Camera Compatible Yes

UART Console UART0
UART Console Baud Rate 115200
Reserved Interfaces UART0, I2C0

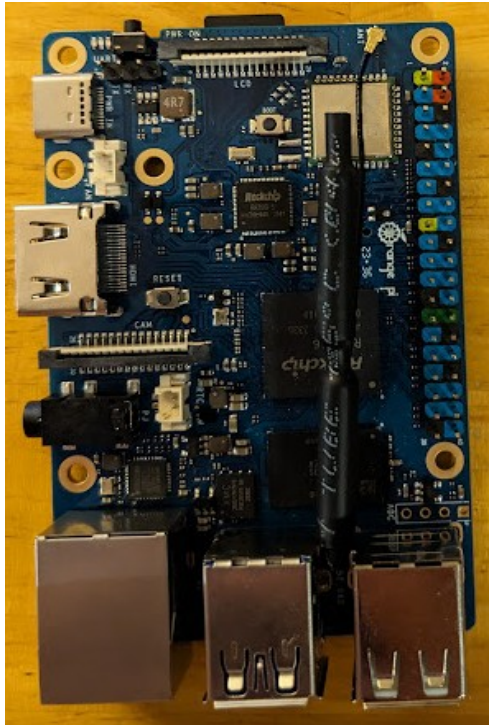
Number USB 4 x USB-A
Powered By USB-C
Boots From micro SD Card

Built-in WiFi Yes – 2.4GHz, 5.0GHz
Ethernet Port Yes

CPU Cores 4
RAM 1GB/2GB/4GB/8GB
GPU Video Decoding Yes

<https://www.raspberrypi.com/documentation/computers/ras>

Orange Pi 3B v1.1.1



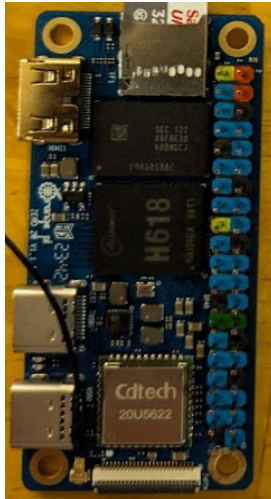
	Physical	
3.3V	1 2	5V
I2C2 SDA	3 4	5V
I2C2 SCL	5 6	GND
GPIO 147	7 8	UART2 TX
GND	9 10	UART2 RX
GPIO 118	11 12	GPIO 119
GPIO 128	13 14	GND
GPIO 130	15 16	GPIO 131
3.3V	17 18	GPIO 129
SPI3 MOSI	19 20	GND
SPI3 MISO	21 22	GPIO 132
SPI3 SCLK	23 24	SPI3 CS0
GND	25 26	SPI3 CS1
I2C3 SDA	27 28	I2C3 SCL
GPIO 133	29 30	GND
GPIO 124	31 32	GPIO 144
GPIO 127	33 34	GND
GPIO 120	35 36	GPIO 125
GPIO 123	37 38	GPIO 122
GND	39 40	GPIO 121

RPi Hat Compatible	Yes – Full 40 pin
Rpi Camera Compatible	No – Hardware might be con
UART Console	UART2
UART Console Baud Rate	1500000
Reserved Interfaces	UART2
Number USB	4 x USB-A
Powered By	USB-C
Boots From	micro SD Card
Builtin WiFi	Yes – 2.4GHz, 5.0GHz
Ethernet Port	Yes
CPU Cores	4
RAM	2GB/4GB/8GB
GPU Video Decoding	No – Not currently supported

<http://www.orangepi.org/html/hardWare/computerAndMic>
http://www.orangepi.org/orangepiwiki/index.php/Orange_F

*It seems v1.1.1 boards use pinout of v2.1 on 40 pin heads
 (physical 7 and 22 were different for v1.1?)*

Orange Pi Zero 2W v1.1.1



	Physical	
3.3V	1 2	5V
I2C2 SDA	3 4	5V
I2C2 SCL	5 6	GND
GPIO 269	7 8	UART0 TX
GND	9 10	UART0 RX
GPIO 226	11 12	GPIO 257
GPIO 227	13 14	GND
GPIO 261	15 16	GPIO 270
3.3V	17 18	GPIO 228
SPI1 MOSI	19 20	GND
SPI1 MISO	21 22	GPIO 262
SPI1 SCLK	23 24	SPI1 CS0
GND	25 26	SPI1 CS1
I2C3 SDA	27 28	I2C3 SCL
GPIO 256	29 30	GND
GPIO 271	31 32	GPIO 267
GPIO 268	33 34	GND
GPIO 258	35 36	GPIO 76
GPIO 272	37 38	GPIO 260
GND	39 40	GPIO 259

RPi Hat Compatible	Yes – Full 40 pin
Rpi Camera Compatible	No – connector on edge is not for cameras

UART Console	UART0
UART Console Baud Rate	115200
Reserved Interfaces	UART2

Number USB	1x USB C (USB1 only)
Powered By	USB-C (either)
Boots From	micro SD Card

Built-in WiFi	Yes – 2.4GHz, 5.0GHz
Ethernet Port	Yes

CPU Cores	4
RAM	1GB/2GB/4GB 1.5GB Version NOT SUPPORTED BY ArPiRobot!!!
GPU Video Decoding	No – Not currently supported in software

<http://www.orangepi.org/html/hardWare/computerAndMicrocontrollers/details/Orange-Pi-Zero-2W.html>
http://www.orangepi.org/orangepiwiki/index.php/Orange_Pi_Zero_2W#How_to_install_wiringOP

Note: I2C bus numbers don't match OrangePi pinout diagrams because they don't show up that way in the OS

Best I can tell, v1.1 and v1.1.1 have the same pinout